

GLYCAN DETERMINANTS OF BIOFILM PERSISTENCE

Neu5Gc, Dietary Innate Agonists, and the Problem of Eradication in Lyme Disease and Related Chronic Conditions

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Sialic-Glycan Inflammatory Cascade (SGIC) Research Program

STATUS OF THIS PAPER

Draft — internal review copy. Not peer reviewed. References marked (V) require verification against source before publication.

This paper follows the SGIC firewall convention: established findings and program hypotheses are presented in visually distinct classes and are never blended. Nothing in this paper constitutes medical advice, and nothing in it should be read as a claim that dietary modification eradicates biofilm. Section 7 addresses that distinction directly, because it is the point at which this argument is most likely to be misused.

ABSTRACT

Biofilm-associated infection is defined clinically by a single stubborn property: the organism is not killed by regimens that reliably kill the same organism in free-living form. This paper examines whether the glycan environment supplied by the host — and specifically the fraction of that environment derived from diet rather than from the human genome — contributes to that failure of resolution.

Three established observations anchor the argument. First, sialic acid is not incidental to biofilm biology but load-bearing: it functions simultaneously as bacterial nutrient, as adhesion substrate once cleaved, and as immune-evasive surface decoration. Second, humans cannot synthesise the sialic acid N-glycolylneuraminic acid (Neu5Gc) yet incorporate it metabolically from red meat and dairy into epithelium, endothelium, and secreted glycans, against which essentially all humans carry circulating antibody. Third, wheat amylase-trypsin inhibitors (ATIs) are characterised TLR4/MD2 agonists, and gliadin fragments modulate epithelial barrier permeability.

From these, four hypotheses are proposed. The central one — the Xenoantigenic Substratum Hypothesis — holds that in chronic biofilm lesions the immunologically foreign glycan is located not within the biofilm but in the host tissue immediately beneath and around it, producing sustained antibody-driven inflammation directed at a target the immune system cannot remove and that does not damage the organism.

A Lyme-specific extension is developed separately, on the grounds that *Borrelia* aggregate matrix is reported as alginate-based rather than sialylated, and therefore the general sialic-acid nutrient argument does not transfer directly. The proposed Lyme mechanism instead runs through N-glycolyl substitution of host glycosaminoglycans and the decorin/heparan-sulfate adhesion axis on which *Borrelia* is known to depend.

Falsification criteria and a low-cost first experiment are specified.

1. SCOPE AND EPISTEMIC STATEMENT

This paper follows the SGIC firewall convention. Three classes of statement appear, and they are never blended:

ESTABLISHED — replicated, peer-reviewed findings from independent laboratories. Cited. Not original to this program.

HYPOTHESIS — proposed by this program. Not established. Presented with explicit falsification criteria.

CONTESTED — published but actively disputed within the field. Used only where the dispute itself is disclosed.

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2. THE CLINICAL PROBLEM

Biofilm-embedded organisms tolerate antimicrobial concentrations orders of magnitude above those effective against planktonic cells of the same strain. This is tolerance, not resistance — it is not primarily genetic, and it does not require the acquisition of resistance determinants.

ESTABLISHED

The mechanisms are well characterised and multiple: incomplete penetration of the extracellular polymeric substance (EPS) matrix; chemical inactivation within the altered microenvironment of the film; metabolic dormancy of persister subpopulations; and the physical exclusion of phagocytes, which cannot ingest a structure larger than themselves and instead undergo frustrated phagocytosis, degranulating onto host tissue.

That final mechanism is where this paper begins. Frustrated phagocytosis means the host's own effector response becomes a source of continuing tissue injury without achieving clearance. Any factor that increases the intensity of that response without increasing its effectiveness makes the clinical picture worse, not better.

The SGIC question is whether dietary xenoantigen is exactly such a factor.

3. PART I — ESTABLISHED SCIENCE

3.1 Sialic acid is structural to biofilm biology

ESTABLISHED

Bacterial sialidases (neuraminidases) cleave terminal sialic acid from host glycoproteins, mucins, and salivary proteins. This serves two purposes at once: it liberates a usable carbon source, and it strips the terminal sugar to expose underlying galactose and GlcNAc residues that serve as adhesion receptors. Sialidase activity is strongly and repeatedly associated with biofilm development across unrelated pathogens. (1, 2)

ESTABLISHED

In *Porphyromonas gingivalis*, sialidase gene expression is substantially higher in biofilm than in planktonic culture, and a sialidase-deficient mutant forms thinner, discontinuous biofilm under acidic, alkaline, and oxidative stress. (2)

ESTABLISHED

In *Tannerella forsythia*, sialic acid acquisition via the NanT transporter is critical during biofilm growth; the organism can substitute sialic acid for N-acetylmuramic acid to sustain growth in the biofilm state. (3)

ESTABLISHED

In cystic fibrosis airways, sialidase-producing microbiota and free sialic acid released from mucin are associated with selection of quorum-sensing mutants and the transition of *Staphylococcus aureus* to a chronic lifestyle. (4)

The pattern is consistent: host sialic acid is a resource, and the enzymatic machinery for taking it is co-regulated with the machinery for persisting.

3.2 Neu5Gc: the dietary xenoantigen

ESTABLISHED

Humans carry an inactivating mutation in CMAH and cannot convert CMP-Neu5Ac to CMP-Neu5Gc. Neu5Gc is nonetheless absorbed from red meat and dairy and metabolically incorporated into human glycoproteins and glycolipids — epithelium, endothelium, placenta, gangliosides, and carcinoma tissue. (5, 6)

ESTABLISHED

Circulating anti-Neu5Gc antibodies of IgG, IgM, and IgA class are present in essentially all humans from approximately six months of age. The combination of incorporated antigen and circulating antibody produces chronic low-grade inflammation, termed xenosialitis. (6, 7)

ESTABLISHED

Neu5Gc supports bacterial growth. *T. forsythia* grown on Neu5Gc as sole carbon source reaches levels comparable to growth on Neu5Ac. Dietary Neu5Gc is, straightforwardly, food for a biofilm-forming periodontal pathogen. (1)

ESTABLISHED

Dietary Neu5Gc creates functional receptors on human tissue. Subtilase cytotoxin B subunit binds preferentially to Neu5Gc-terminating glycans; assimilation of dietary Neu5Gc generates high-affinity toxin receptors on human gut epithelium and kidney vasculature that would otherwise not exist. This is a receptor manufactured by diet, not by genome. (8)

ESTABLISHED

Red-meat-rich diets shift the gut microbiome toward greater sialidase production. (9)

3.3 The counterintuitive finding, and why it matters most

ESTABLISHED

Nontypeable *Haemophilus influenzae* preferentially incorporates Neu5Ac over Neu5Gc into its lipooligosaccharide by roughly fourfold, and its CMP-sialic acid synthetase (SiaB) shows approximately 4,000-fold higher catalytic efficiency for Neu5Ac. The interpretation offered is that this represents host adaptation under selective pressure from the human anti-Neu5Gc antibody response. (10)

This finding is frequently read as an objection to Neu5Gc pathology in infection. It should be read the opposite way, and this paper's central hypothesis follows directly from it.

The bacterium has been forced to avoid displaying Neu5Gc. That is evidence the anti-Neu5Gc response works — well enough to shape bacterial genome evolution. It also relocates the question. If the organism has scrubbed Neu5Gc from its own surface, the remaining Neu5Gc in a chronic lesion is on the host.

3.4 ATIs and gliadins: the innate terrain

ESTABLISHED

Wheat amylase-trypsin inhibitors activate the TLR4–MD2–CD14 complex on intestinal myeloid cells, driving release of inflammatory mediators and promoting intestinal inflammation. This is a nutritional, non-coeliac, non-IgE innate activation pathway. (11, 12)

ESTABLISHED / PARTLY CONTESTED

Gliadin fragments trigger zonulin release and increase paracellular intestinal permeability. The phenomenon is reproducible; its magnitude and clinical significance outside coeliac disease remain debated. (13)

Neither ATIs nor gliadins act on biofilm. They act on terrain: mucin quantity and glycosylation, secretory IgA output, antimicrobial peptide production, and the baseline activation state of circulating myeloid cells. That distinction is preserved throughout this paper.

3.5 *Borrelia* biofilm: what is established and what is disputed

This section is deliberately conservative. Overstatement here is the most likely reason for the paper to be dismissed unread.

ESTABLISHED (IN VITRO)

B. burgdorferi forms aggregates in vitro displaying recognised biofilm hallmarks: staged structural rearrangement, development on abiotic and biotic substrates, internal channels and surface protrusions on atomic force microscopy, and secretion of a protective EPS matrix. Matrix analysis identified sulfated and non-sulfated/carboxylated substrates, predominantly alginate, with calcium and extracellular DNA. (14)

ESTABLISHED (IN VITRO)

Stationary-phase and aggregate forms of *B. burgdorferi* survive antibiotic exposure at markedly higher rates than motile spirochetes, with substantially elevated minimum biofilm inhibitory concentrations. (15)

CONTESTED

Reports of *Borrelia* biofilm in vivo — in borrelial lymphocytoma skin biopsies and in infected mouse heart tissue — come predominantly from a single research group, and the field does not treat the in vivo case as settled. Independent replication is limited, and the broader question of whether viable organisms persist after treatment remains one of the most contentious in infectious disease. In the murine model, *B. burgdorferi* DNA and antigen have been shown to persist post-treatment without demonstrated viable culturable organisms — a finding compatible with several competing explanations. (16, 17)

POSITION TAKEN HERE

This paper does not assume that persistent viable *Borrelia* biofilm is the cause of post-treatment Lyme disease syndrome (PTLDS). It argues something weaker and more defensible: that retained antigen in a matrix-protected niche — whether that matrix contains viable organisms, dead organisms, or debris alone — is sufficient to sustain the inflammatory picture described below, provided the surrounding host tissue is itself antigenically marked. The hypothesis does not require the persistence debate to be resolved in either direction.

4. PART II — SGIC HYPOTHESES

H1 · The Xenoantigenic Substratum Hypothesis (central)

HYPOTHESIS

In chronic biofilm-associated lesions, dietary Neu5Gc incorporated into host epithelial, endothelial, and matrix glycans beneath and adjacent to the film constitutes a fixed, non-clearable antigenic target. Anti-Neu5Gc antibody deposition at this site recruits complement and effector cells into a compartment where the actual organism is physically inaccessible. The result is sustained, localised, antibody-driven inflammation that damages host tissue, degrades barrier integrity, and liberates further glycan substrate — without exerting meaningful pressure on the organism.

The lesion therefore does not resolve, and the failure to resolve is not primarily a failure of antimicrobial potency. It is a target-selection failure.

Corollary. Because the organism has evolved to avoid Neu5Gc display (§3.3), the higher the host's Neu5Gc burden, the more sharply the antibody response is misdirected toward self tissue rather than toward the pathogen. Dietary Neu5Gc does not merely add inflammation — it degrades the discrimination of an otherwise functional response. This is the self/non-self axis of the SGIC framework applied to a defended structure.

H2 · The Sialidase Selection Hypothesis

HYPOTHESIS

Habitual dietary Neu5Gc and animal-sialoglycoprotein load selects, at mucosal sites, for sialidase-competent and sialic-acid-catabolising taxa (established, §3.1–3.2). Since sialidase is co-regulated with biofilm formation in several pathogens, this constitutes dietary selection for a biofilm-competent microbial community. The effect is ecological and cumulative rather than acute, and is predicted to be strongest at oral, sinonasal, urogenital, and gastrointestinal mucosae.

H3 · The N-Glycolyl Glycosaminoglycan Hypothesis (Lyme-specific)

This is the hypothesis this paper considers most novel and most worth testing.

The general sialic-acid nutrient argument transfers poorly to *Borrelia*, whose reported matrix is alginate-based rather than sialylated (§3.5). A different route is required.

ESTABLISHED

B. burgdorferi adhesion to mammalian tissue depends heavily on host glycosaminoglycans. Decorin-binding proteins A and B (DbpA/DbpB) mediate attachment to decorin and to dermatan sulfate; DbpA binding is abolished by mutation of lysines K82, K163, K170, and that in vitro defect translates to a defect during murine infection. DbpA is highly polymorphic between strains, and allelic variation in DbpA measurably shifts tissue tropism. Spirochetes disseminate less efficiently in decorin-deficient mice. (18, 19, 20)

ESTABLISHED — AND DIRECTLY LOAD-BEARING FOR THIS PAPER

Decorin's glycosaminoglycan chain is preferentially dermatan sulfate in skin and chondroitin sulfate in cartilage. (21)

ESTABLISHED — THE CRITICAL FINDING

In ceftriaxone-treated mice, *B. burgdorferi* DNA was detected in joint tissue uniformly in animals infected with DbpA/B-expressing bacteria, but not in animals infected with a DbpA/B-deficient strain. All treated animals were culture negative, and remained so under anti-TNF- α immunosuppression, indicating that the persisting material is DNA or DNA-containing remnant rather than viable organism. The authors further note that decorin-rich tissues such as joint may serve as a protective niche in which the organism is shielded from the immune response. (22)

This last result is the single most important citation in this paper, and it comes from a group independent of the *Borrelia* biofilm literature. It establishes, without requiring any contested premise, that:

- post-treatment persistence of borrelial material in joint is GAG-adhesion dependent;
- the persisting material is antigen, not viable organism;
- the niche in which it persists is decorin-rich connective tissue.

The SGIC framework needs nothing more than this to proceed. The question it adds is simply: what else is antigenic in that niche?

ESTABLISHED

Catabolism of Neu5Gc results in transfer of the N-glycolyl group into glycosaminoglycans. N-glycolyl chondroitin (Gc-CN) has been detected in human tissue and in several mammalian species. (23)

HYPOTHESIS

Dietary Neu5Gc alters the fine chemistry of host dermatan/chondroitin sulfate — the precise substrate class on which *Borrelia* tissue tropism depends. Two consequences are predicted, and they are separable:

H3a (adhesion). N-glycolyl substitution modifies DbpA/DbpB binding affinity, altering the efficiency of colonisation and the anatomical distribution of organism burden.

H3b (antigenicity). N-glycolyl-substituted GAG in skin and joint capsule constitutes an anti-Neu5Gc antibody target at exactly the sites of *Borrelia* tropism and of the dominant PTLDS symptom complex — skin, joint, connective tissue. The inflammation is then directed at connective tissue rather than at the organism.

H3b, if supported, would offer a mechanism for the most clinically frustrating feature of PTLDS: symptoms concentrated in connective tissue, persisting after antimicrobial therapy, with inflammatory markers that do not track organism burden.

H4 · The Innate Priming Hypothesis

HYPOTHESIS

Chronic ATI-driven TLR4 activation and gliadin-associated barrier permeability raise the baseline activation state of the myeloid compartment and degrade mucosal barrier quality, lowering the threshold at which a marginal biofilm lesion becomes a chronic one. ATIs and gliadins are proposed as permissive, not causal — they do not create the film, they degrade the conditions under which the host would otherwise contain it.

5. PREDICTIONS AND FALSIFICATION CRITERIA

The framework is falsifiable. It is wrong if:

#	PREDICTION	FALSIFIED IF
P1	Neu5Gc is detectable, and enriched relative to uninvolved tissue, in host tissue immediately subjacent to persistent biofilm	Subjacent tissue shows Neu5Gc at or below matched uninvolved control levels
P2	Immunoglobulin and complement deposition co-localises with Neu5Gc-positive host tissue at the lesion margin	Deposition is spatially unrelated to Neu5Gc distribution
P3	Neu5Gc tissue burden correlates with lesion chronicity, not with acute severity	No correlation, or correlation with acute severity only
P4	N-glycolyl-substituted GAG is detectable in human dermis and joint capsule and varies with dietary intake	Gc-GAG absent from these tissues, or invariant across dietary exposure
P5	Anti-Neu5Gc titre falls measurably over 6–12 months of strict elimination	Titres unchanged under verified adherence
P6	Symptom improvement under elimination, where it occurs, precedes any change in organism burden	Improvement tracks organism burden — indicating an antimicrobial rather than immunological mechanism

P6 is the discriminating prediction. This framework claims elimination works by withdrawing a misdirected target, not by killing anything. If improvement tracks organism clearance, the framework is describing the wrong mechanism, even if the clinical observation holds.

6. PROPOSED FIRST EXPERIMENT

The entry experiment is cheap, uses existing reagents, and yields a publishable result in either direction.

Design. Immunohistochemical detection of Neu5Gc in archived and prospectively collected tissue from biofilm-associated lesions, with matched uninvolved tissue from the same patient as internal control.

Reagent. Chicken IgY polyclonal anti-Neu5Gc, which recognises Neu5Gc across a broader range of underlying structural contexts than available monoclonals, with the established sialidase-pretreatment specificity control. (24)

Tissue targets, in ascending order of access difficulty:

1. Explanted continuous glucose monitor sensors and surrounding capsule (links directly to the SGIC whitepaper on CGM foreign body response; large accessible supply; minimal ethics burden)
2. Chronic wound slough and debridement margin
3. Chronic rhinosinusitis polyp tissue
4. Periodontal plaque and adjacent gingival tissue
5. Archived borrelial lymphocytoma / erythema migrans biopsy material — the direct Lyme test, and the hardest to obtain

Co-stain. IgG and C3d, to test P2.

Result interpretation. Enrichment of Neu5Gc in lesional host tissue relative to matched control, with co-localised immunoglobulin deposition, converts H1 from hypothesis to finding. Absence of enrichment falsifies the central claim of this paper cleanly, and that result should be published too.

7. WHAT THIS PAPER DOES NOT CLAIM

SCOPE LIMITATION — NOT TO BE REMOVED IN DERIVATIVE MATERIAL

This section is not a disclaimer. It is a substantive limitation of scope, and it should not be removed in summary or derivative material.

Dietary elimination is not proposed as a biofilm eradication strategy. No published evidence demonstrates that removal of Neu5Gc, ATIs, or gliadins reduces biofilm burden or accelerates clearance in any condition. The mechanisms of biofilm antibiotic tolerance listed in §2 — matrix diffusion limitation, persister dormancy, altered microenvironment — are unaffected by diet and are, on current evidence, the dominant determinants of eradication failure. A mature biofilm is not dissolved by a dietary change.

The claim is narrower and different in kind. Removal of dietary xenoantigen is proposed to withdraw a misdirected antigenic target, reduce a chronic innate activation load, and thereby return immune

capacity and tissue-repair capacity that were being consumed unproductively. In the clinical language: it removes a handicap. It is not a solvent.

Attribution is confounded. Eliminating red meat, dairy, and wheat changes fibre intake, fat composition, fermentable substrate, and microbiome composition simultaneously. Symptomatic improvement under such a change cannot be attributed to Neu5Gc, ATI, or gliadin specifically without controls that do not currently exist. This is the single largest methodological weakness in the observational base underlying the SGIC framework, and it is stated here rather than left for a reviewer to state.

Competing explanations for PTLDS remain viable and are not displaced by this paper: post-infectious autoimmunity independent of dietary glycan, residual antigen without viable organism, tissue damage sustained during acute infection, central sensitisation, and misdiagnosis of an unrelated condition. This framework is compatible with several of these rather than competitive with them.

8. CONCLUSION

Biofilm defeats clearance by making the organism physically unreachable. This paper proposes that in the human host, a second and separable problem operates alongside the first: the immune system, denied access to the organism, engages the only antigenically foreign material it can reach — which, in a Neu5Gc-loaded host, is the host's own tissue.

If that is correct, chronic biofilm-associated disease in humans is partly a disorder of target selection rather than solely of antimicrobial access, and the fraction attributable to target selection is dietary in origin and therefore modifiable.

The claim is testable with existing reagents on existing tissue. It should be tested.

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